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A system factorial technology analysis of the size congruity effect: Implications for numerical cognition and stochastic modeling

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HIGHLIGHTS

- The SCE is the phenomenon by which numerical magnitude interferes with or facilitates physical judgments of numerals.
- The SCE has been interpreted as a strong evidence for automatic activation of numerical magnitude.
- We have applied the system factorial technology (SFT, Townsend & Nozawa, 1995) – a stochastic modeling approach – to unravel the underlying mechanisms of the SCE.
- We found substantial evidence for serial minimum-time processing — an architecture that is inconsistent with automatic activation.
- We conclude that numerical information is not activated in an automatic fashion.
- We explain how the SCE can be present or absent depending on the specific model of serial minimum-time models.

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ABSTRACT

We applied the methodology known as the system factorial technology (SFT) to diagnose the information-processing architecture underlying the size-congruity effect (SCE) in numerical cognition. The SCE documents the interference in judging the *physical* size of numerals when this size disagrees with their numerical magnitude or the facilitation when the two attributes agree. Traditional theories of the SCE implicate the automatic activation of numerical magnitude and hence the mandatory interaction in processing between number and size. In contrast, in a pair of experiments we found serial minimum-time processing of number and size, an outcome which excludes the possibility of interaction. In the face of this architecture, we still recorded appreciable amounts of redundancy gains when number and size corresponded (=SCE). However, we show that this SCE does not derive from an interaction in processing. We show that, given stochastic independence, certain species of serial self-terminating models actually mandate the SCE. Other species of serial self-terminating models do not allow an SCE, an outcome that accounts for the absence of an observable SCE in a fair number of studies. Our results are inconsistent with the belief that numerical information is activated in an automatic fashion under all circumstances.

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In (pure) mathematics, numbers are dimension-less objects. In the empirical world, by contrast, numbers come dressed in physical dimensions. The numbers in everyday life are written on classroom blackboards, appear on computer screens, or sounded by a teacher. In the absence of a physical/sensory dimension, no human would have ever perceived a number. Given that each number is a combination of semantic and physical attributes, one can ask: How do people perceive the *physical* dimension of the number? In particular, does numerical magnitude affect perception of the digit's physical size? Many studies (see, Dehaene, 1997, for review) show that it does, which, in turn, led researchers to argue that the arithmetic/semantic dimension – numerical magnitude – is processed in an automatic fashion just whenever a numeral is presented for view for any purpose. Given the automatic processing of number, a number–physical size interaction is inevitable whenever different numbers in different physical sizes are processed (especially in tasks referring to physical size). In this study, we subjected the assumption of automatic number processing to new scrutiny, harnessing state-of-the-art tools of stochastic modeling. To anticipate our conclusions, we show that the architecture governing processing excludes the possibility of a number–size interaction. Consequently, we challenge the assumption of automatic processing of numerical magnitude.

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1. The size congruity effect

When presented with a pair of numerals that differ in physical size, people select the *physically* larger member of the pair faster when this member is also numerically larger than when it is numerically smaller. Thus, the selection of the larger format is faster in the pair, 7-2, than in the pair, 2-7. The dimensional values match in the first pair (a congruent display), but conflict in the second pair (an incongruent display). The difference in performance between congruent and incongruent trials (favoring the former) defines the size congruity effect (SCE). One does not need to present pairs of numerals in order to observe the SCE. When presented with a single numeral, it takes people longer to decide that the physical format is small when the numeral is 9 than when the numeral is 2 (Algom, Dekel, & Pansky, 1996; Choplin & Logan, 2005; Fitousi, 2014). The SCE has been obtained with numbers written in different notations and in different tasks (e.g., Besner & Coltheart, 1979; Fitousi, 2010; Ito & Hatta, 2003; Schwarz & Ischebeck, 2003), with natural and negative numbers (Henik & Tzelgov, 1982; Pansky & Algom, 1999, 2002; Pinhas & Tzelgov, 2012; Pinhas, Tzelgov, & Guata-Yaakobi, 2010; Schwarz & Heinze, 1998; Szücs & Soltész, 2008; Tzelgov, Ganor-Stern, & Maymon-Schreiber, 2009; Tzelgov, Meyer, & Henik, 1992), and with single- and double-digit numbers (Fitousi & Algom, 2006; Fitousi, Shaki, & Algom, 2009; Ganor-Stern, Tzelgov, & Ellenbogen, 2007; Santens & Verguts, 2011). Given its pervasiveness, the SCE is commonly taken to reflect on the mandatory processing of numerical magnitude. It is this view that we challenge in the current study.

2. Is the SCE mandatory?

The impressive evidence notwithstanding, pervasive biases in the standard experimental design call into question the obligatory nature of the SCE. Little stimulus alchemy suffices to eliminate the SCE or reverse the effect such that physical size intrudes on perception of numerical magnitude more than vice versa (the reverse-SCE; Fitousi & Algom, 2006). Algom et al. (1996) identified two critical biases prevalent in published SCE studies. First, there is a glaring asymmetry in the number of stimuli used for the numerical and the physical dimensions. Typically, the numbers 1 to 9 (inclusive) are used for the former, but only two or three values (small, medium, large) are used for the latter. Virtually all pertinent research pitted a finely grained numerical dimension against a coarse physical dimension. This asymmetry itself can determine the observed interaction (=SCE). Melara and Mounts (1994) have shown that the mere number of stimuli on an irrelevant dimension affects classification performance on the relevant dimension (see also Melara & Algom, 2003; Sabri, Melara, & Algom, 2001).

For another bias, the relative salience or discriminability of values along the number and the size dimensions was not matched. Discriminability is matched if the values along the numerical dimension differ perceptually from one another to the same extent as do values of size along the physical dimension. This information is critical because the more discriminable dimension can disrupt performance on the less discriminable dimension (e.g., Algom et al., 1996; Fitousi & Algom, 2006; Melara & Mounts, 1994). Thus, irrelevant numerical value will disrupt performance with physical size (=SCE) if the values of number differ psychologically from one another more than do the values of physical size from one another. Mismatched discriminability favoring numbers was present in virtually all studies of the SCE. However, when care was taken to match discriminability (and number of values along the two dimensions), the SCE collapsed. And, when physical size was purposely made more salient than numerical value, a reverse SCE emerged (Algom et al., 1996; Fitousi, 2014; Fitousi & Algom, 2006; Fitousi et al., 2009; Pansky & Algom, 1999, 2002; see also, Melara & Algom, 2003; Sabri et al., 2001). In Fig. 1, we illustrate the effect on the SCE of relative dimensional salience.

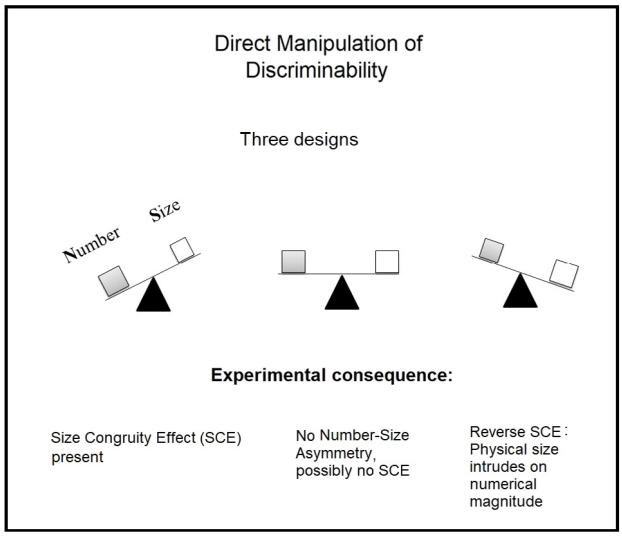


Fig. 1. Schematics of the influence of relative dimensional salience on the outcome of the SCE experiment. Left-hand panel: The numbers (N) are more discriminable than the values of physical size (S), the default setup in many studies. As a result, the irrelevant numbers intrude on the judgments of size, thereby generating the SCE. Middle panel: Numerical magnitude and physical size are matched in discriminability, resulting in the elimination of the SCE asymmetry in interference favoring numbers. Right-hand panel: Physical size is more discriminable than numerical magnitude, so that judgments of numbers are now subject to interference from physical size more than vice-versa (=reverse SCE). We uncover through SFT-guided collection of data and modeling the architecture underlying each of the outcomes depicted in Fig. 1 (see Discussion).

The malleability of the SCE casts doubt on the automatic nature of processing numerical magnitude. The question of the mandatory nature of the SCE is related to the fundamental contrast in cognitive science drawn by Garner (1970, 1974) between integral and separable dimensions (Garner, 1970, 1974, 1976; Garner & Felfoldy, 1970; see Algom & Fitousi, 2016, for a review). Separable dimensions of an object do not intrude on one another in processing, whereas integral dimensions do — due to holistic perception of the object. The elimination of the SCE in the Algom studies is consistent with the view that numerical magnitude and physical size of numbers are separable dimensions, whereas the robust appearance of the SCE is consistent with the traditional view that number and size are integral dimensions.

3. Are dimensions of number integral or separable?

All objects of perception are not experienced in the same way. Some show a character of totality, a *Gestalt* (cf. Heidbreder, 1933, p. 331), whereas other objects are experienced as separate features, which are simply grouped together in space and time. The objects in the first class are composed of integral dimensions, whereas the objects in the second class are composed of separable dimensions. Integral dimensions are processed in a unitary fashion, so that observers cannot attend exclusively to one dimension of the object without concurrently noticing the other dimension. It is the cross-talk between the constituent dimensions that yields the observed interaction in performance. Prototypical examples of integral dimensions are hue, brightness, and saturations of colors. According to the traditional view (Dehaene, 1997), numerical magnitude and physical size form another example of integral dimensions, expressed by the SCE.

Separable dimensions, by contrast, remain distinct in processing. Observers can successfully ignore a task-irrelevant dimension

when attending to the target dimension. Prototypical examples of separable dimensions are color and shape of an object or size and angle of diameter of circles. Given independent processing with separable dimensions, performance with each does not yield interaction (facilitation, interference). The studies by Algom and his associates can be taken to show that numerical value and physical size are separable dimensions.

The assumption of automatic activation of numerical value is consistent with integral processing because magnitude is always activated and intrudes on physical size. However, the presence of an SCE may result from identifiable biases in the typical experimental design, not from truly integral processing. Consequently, the presence of the SCE does not, by itself, imply an interactive architecture. It is at this juncture that application of stochastic modeling is invited. The explication of the underlying architecture that sustains the SCE is vital for the understanding of its source and, and in general, for our understanding of number. We elected to deploy the tools of stochastic modeling subsumed under the umbrella of system factorial technology (SFT; Townsend & Nozawa, 1995) in order to uncover the system underlying processing of the SCE in numerical cognition.

4. A synopsis of SFT applied to numerical processing

Four distinct yet interrelated features of performance are specified in the SFT: (1) Architecture (serial vs. parallel, and, if the latter, coactive or not); (2) Stopping rule (minimum time vs. exhaustive); (3) Capacity (limited vs. unlimited vs. super), and (4) Stochastic independence (independence vs. dependence). The SFT has been serving mathematical psychologists as a tool for theoretical resolution in a variety of simple visual detection or memory-search tasks (e.g., Fific, Little, & Nosofsky, 2010; Fific, Nosofsky, & Townsend, 2008; Fitousi, 2015; Townsend & Wenger, 2004). In this study, the task was not simple visual detection, but a decision with respect to numbers (see Algom, Fitousi, & Eidels, 2017). The participants decided on each trial whether the number was larger or smaller numerically than a specified boundary or whether its physical size was larger or smaller than a criterial physical format.

We made use of the SFT version known as the *double factorial paradigm* (DFP; Townsend & Nozawa, 1995; see also, Algom, Eidels, Hawkins, Jefferson, & Townsend, 2015; Eidels, Townsend, & Algom, 2010). We illustrate the paradigm as it applies to the current stimuli in Fig. 1. Our stimuli varied along numerical magnitude (large, small) and physical size (large, small) of the presented digits. The binary values of the two dimensions were combined to form a factorial design. The target was defined as a large stimulus on either dimension or both. Therefore, the observer was to respond YES if the numerical value (N) was large (i.e., $N > 5$), if the physical format (S) was large ($S >$ 2 cm in height), or if both were large (i.e., $N > 5$ AND $S >$ 2 cm in height). Otherwise the observer was to respond NO.

A notable aspect of the DFP is that a second variable, salience, is also manipulated along each dimension. For the dimension of numerical magnitude, the target number could be salient, namely, noticeably larger than the boundary of 5 (i.e., 9) or less salient and closer to the boundary (6). This manipulation of numerical salience makes use of the well documented distance effect (Fitousi & Algom, 2006; Moyer and Landauer, 1967). For the dimension of physical size, the format could be salient, namely, noticeably larger than the boundary of 2 cm (6 cm) or less salient and closer to the boundary (4 cm). For trials including targets on both dimensions (hence double-target trials), salience creates a second factorial design (salience of the target magnitude: low, high; salience of the target size: low, high).

It is assumed that the two manipulations of salience *selectively* influence the respective processing of numerical magnitude and of physical size (see e.g., Sternberg, 1969; Townsend & Nozawa, 1995; Van Zandt, 2000, for the critically important notion of selective influence). This means faster and more efficient processing of salient targets than less-salient targets on each dimension. Moreover, rendering numerical magnitude salient does not affect the processing of physical size and, conversely, presenting a salient physical size does not affect the processing of numerical magnitude. We test for selective influence via the survivor ordering condition, but note another approach developed by Dzhafarov and his colleagues (e.g., Dzhafarov, 2003; Dzhafarov, Schweickert, & Sung, 2004).

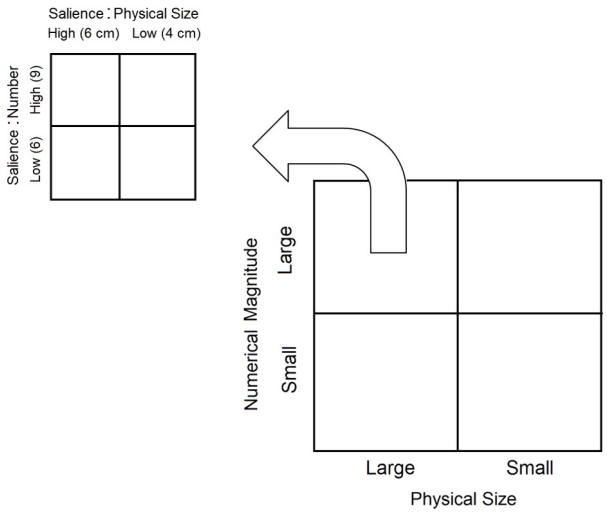


Fig. 2. The DFP applied to the dimensions of Numerical Magnitude and Physical Size of single digits. With 'largeness' defined as target, the factorial design at the bottom crosses the binary values of the two dimensions, creating double-target, single-target, and no-target displays. The concurrent manipulation of salience creates the second factorial design at the top, based on the subset of double-target presentations. The values in parentheses give the values of salient and less-salient targets in each dimension. Fig. 2 refers to the stimuli used in Experiment 1. In Experiment 2, the dimensions of number and line length were used within the same DFP design (see Method for some minor differences between the experiments).

4.1. Mean interaction contrast

A key data of interest are the trials in which both targets are present (the factorial design at the upper left of Fig. 2). There are four possible target combinations, LL, LH, HL, and HH, with L referring to low salience and H to high salience. Assuming the presence of selective influence, namely that salient stimuli lead to faster responding than do less salient stimuli on each dimension, the factorial plots in Fig. 3 present five different possible patterns for the mean RTs. The mean RT patterns can be summarized by first calculating the vertical distance between the leftmost and rightmost points of each line and then calculating the difference between these distances. This double difference is known as the mean interaction contrast (MIC):

$$\begin{aligned} \text{MIC} &= [\text{MRT}_{\text{LL}}(t) - \text{MRT}_{\text{LH}}(t)] - [\text{MRT}_{\text{HL}}(t) - \text{MRT}_{\text{HH}}(t)] \\ &= \text{MRT}_{\text{LL}}(t) - \text{MRT}_{\text{LH}}(t) - \text{MRT}_{\text{HL}}(t) + \text{MRT}_{\text{HH}}(t). \end{aligned}$$

Clearly, MIC $= 0$ when there is no interaction, otherwise the interaction yields an MIC that is different from zero. The MIC is positive for the interaction labeled "over-additive" and it is negative for the interaction labeled "under-additive".

The factorial plots in Fig. 3 provide clues concerning the pertinent processing architecture and stopping rule. If processing is strictly serial, the MIC equals zero regardless of the operative stopping rule (self-terminating or exhaustive). In contrast, parallel

Table 1
Predictions of response–time models in SFT.

Architecture	Stopping rule	Predictions
Parallel	Minimum time	MIC > 0; SIC$(t) > 0$
Parallel	Exhaustive	MIC < 0; SIC$(t) < 0$
Serial	Minimum time	MIC $= 0$; SIC$(t) = 0$
Serial	Exhaustive	MIC $= 0$; SIC$(t) < 0$ for small t, > 0 for large t
Coactivation		MIC > 0; SIC$(t) < 0$ for small t, > 0 for large t

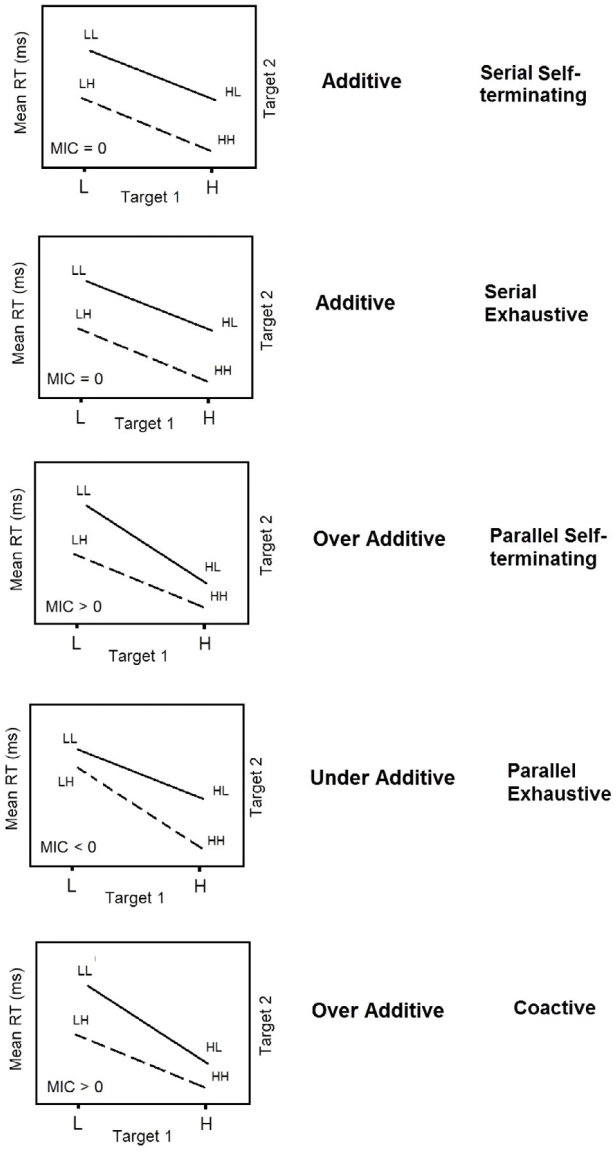


Fig. 3. Illustration of patterns of MIC. For serial processing, the MIC is zero regardless of the operative stopping rule. For parallel processing, the MIC is positive for a minimum time stopping rule, but negative for a maximum time stopping rule. Note that the same pattern (''over-additive interaction'') obtains for parallel minimum time processing and for parallel-coactive processing in which the channels coalesce downstream.

self-terminating processing and parallel coactive processing both yield an over-additive patterns. The reason is that one expects fast responding when either or both dimensions entail a salient target. The single combination, LL, yielding slower responses, is what produces the over-additive pattern. Finally, parallel exhaustive processing results in an under-additive pattern. Under this regime, RT is determined by the slower of the two dimensions, which means that performance will be fast only for the HH combination. These intuitions are supported by formal proofs (Townsend & Nozawa, 1995), and the main results are summarized in Table 1.

It can be seen in Table 1 that the MIC does not distinguish between serial exhaustive and serial self-terminating processing. It also does not distinguish between parallel self-terminating processing and parallel coactive processing. However, the diagnostic power is increased when one considers the entire distribution rather than the means.

4.2. Survivor interaction contrast

Suppose that the entire range of RTs is subdivided into small time bins and that a factorial plot similar to those in Fig. 3 is calculated for each time bin. For sake of mathematical simplicity, the same calculation is made with respect to the values of the survival functions in each bin, rather than with respect to the mean RTs. The survival function, S, for a random variable, T (corresponding to time of processing), is the probability that T takes longer to complete than t, so that $S(t) = p(T > t)$. Note that $S(t) = 1 - F(t)$, with $F(t)$ denoting the distribution function for the same process. In a manner analogous to the calculation of the factorial plot for the mean RTs, one calculates in each time bin, t, the values of the survival function associated with each of the four types of target combinations, $S_{LL}(t)$, $S_{LH}(t)$, $S_{HL}(t)$, and $S_{HH}(t)$. The quantity known as the survivor interaction contrast (SIC) is then given by:

$$SIC(t) = [S_{LL}(t) - S_{LH}(t)] - [S_{HL}(t) - S_{HH}(t)] = S_{LL}(t) - S_{LH}(t) - S_{HL}(t) + S_{HH}(t).$$

Considering Table 1 again, applying the SIC(t) function yields distinctive predictions with respect to serial processing as well as with respect to parallel self-terminating and parallel coactive processing. The SIC(t) function yields fine-grain information, although the fact that the MIC is the integral of the SIC(t) puts constrains on the possible distribution of areas under the SIC(t) curve. For serial self-terminating processing, the SIC(t) is equal to zero at all time, t. In contrast, for serial exhaustive processing, the SIC(t) is negative at early values of t, but positive at latter values. Notably, the total area spanned by the negative portion should equal that by the positive portion. The two species of parallel processing are also distinct now. The SIC(t) is positive at all values, t, for self-terminating parallel processing, but it changes from an early moderate negativity to a predominant positivity at later values of t. The behavior of SIC(t) in the different models is illustrated in Fig. 4.

4.3. Capacity: Miller's inequality

The basic factorial design (at the bottom of Fig. 2) is well suited to gauge capacity. Is performance better with double-target displays than with single-target displays? If so, does the redundant-target advantage indicate dependence in processing of the two targets? These questions were first pursued via an RT inequality suggested by Miller (1978, 1982), and known as the race model inequality,

$$F_{NS}(t) \leq F_N(t) + F_S(t),$$

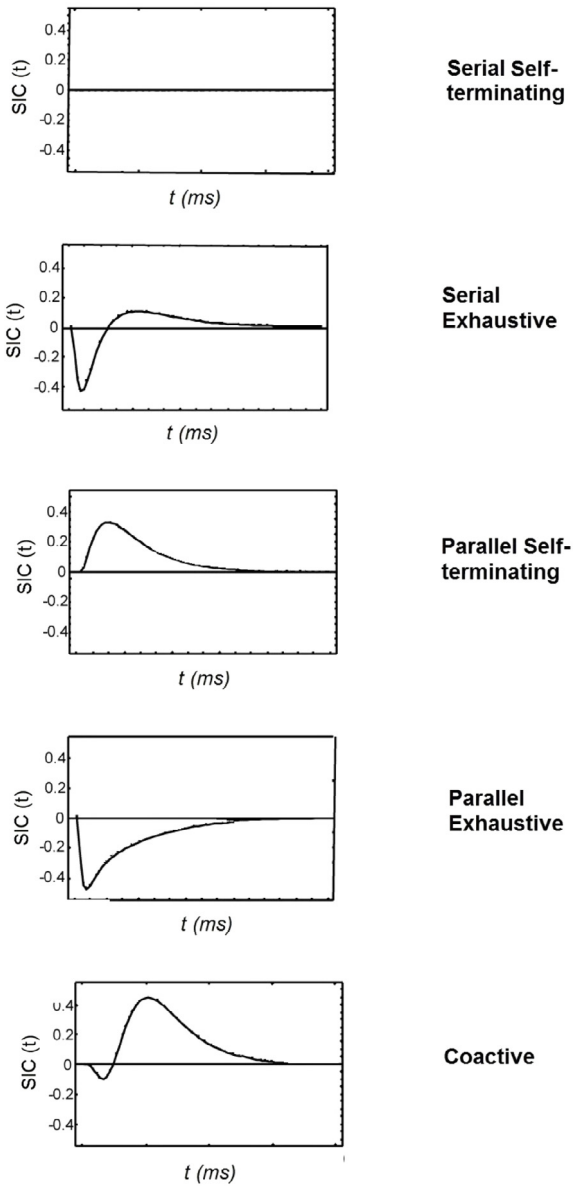


Fig. 4. Illustrative SIC(t) functions for the different architectures and stopping rules. Note in particular the diagnosis provided for the two serial models and for the parallel self-terminating and the parallel coactive models.

in which *F(t)* is, again, the distribution function for the number-target only (N), the size-target only (S) and the double-target (NS) presentations. According to the inequality, the distribution function for the double-target trials cannot exceed the sum of the single-target distribution functions — if indeed number and size are independent in processing (if there is a parallel race without any cross-talk between the two channels). Consequently, violations of the inequality have been interpreted to reject parallel race in favor of co-activation.

However, a double-target advantage can still emerge when the race model inequality is not violated although its magnitude would be fairly limited (Biederman & Checkosky, 1970; Raab, 1962). For another caveat, Ashby and Townsend (1986) and Luce (1986) have independently noticed that the Miller prediction depends on a critical assumption, *context invariance*, namely, that the time to process a target is invariant in single or joint presentations (Colonius, 1990). Let $P_N(T_N \leq t)$ and $P_S(T_S \leq t)$ be the distribution functions for processing each target alone. Let $P_{NS}(T_N \leq t)$ and $P_{NS}(T_S \leq t)$ be the marginal distribution functions for processing the same targets presented with the other target. If context invariance holds, then $P_N(T_N \leq t) = P_{NS}(T_N \leq t)$ and $P_S(T_S \leq t) = P_{NS}(T_S \leq t)$. It is not clear if context invariance holds in our (and other) experimental situations and it is similarly unclear how to measure this critical stipulation (Luce, 1986; Townsend & Honey, 2007; Townsend & Wenger, 2004).[1]

4.4. Capacity: The capacity coefficient

The Miller inequality is recast in the SFT as a statement about the capacity of the system. If the rate of processing a target is slowed down when presented along with another target, the system is of *limited capacity* (there is not a double-target advantage). When processing a target is the same in single and in joint presentations, the system is of *unlimited capacity*. Notably, if the presence of two targets speeds up the ability of the system to process each target, the system is *supercapacity*. Townsend and Nozawa (1995) proved that when the Miller inequality is violated, the system is supercapacity (at that t). The capacity of the observer's system is captured formally by the capacity coefficient, $C(t)$,

$$C(t) = \frac{\log[S_{NS}(t)]}{\log[S_N(t)\,S_S(t)]}.$$

The survivor function is applied because it makes the pertinent development simpler. When $C(t) < 1$, the system is limited capacity, whereas $C(t) = 1$ attests to unlimited capacity. Supercapacity is indicated by $C(t) > 1$. We note that another capacity equation exists for an AND design (not used in our study).

4.5. Capacity: Grice's inequality

The Miller inequality sets the upper bound on performance with double-target displays. There is another bound, proposed by Grice and his colleagues (Grice, Canham, & Boroughs, 1984), that provides a lower bound on RT:

$$F_{NS}(t) \geq MAX[F_N(t), F_S(t)].$$

To satisfy the Grice bound, performance with the faster of the single targets cannot exceed that obtained with the double-target displays. If this nonetheless happens, then the system is of limited capacity to a rather serious degree. Not only is the double-target advantage absent, but also the violation attests to the strain created by increased workload (i.e., by the addition of another target). Again, another inequality captures the bound in an AND design.

4.6. Unified space for capacity measures

The SFT capacity coefficient, the Miller inequality, and the Grice inequality all aim at gauging capacity. However, they are defined on different scales, which has rendered comparison challenging. Townsend and Eidels (2011) have developed a unified framework that allows one to present the three measures within the same space. Miller's inequality can be rearranged in terms of the capacity coefficient as follows,

$$C(t) \leq \frac{\log[S_N(t) + S_S(t) - 1]}{\log[S_N(t)\,S_S(t)]}$$

[1] Sometimes the term context independence is used (Townsend & Nozawa, 1997), carrying the same meaning as context invariance. We note that several procedures under various designations have been developed to capture the core notion of independence in processing (see especially Fitousi, 2013; Fitousi & Wenger, 2013; Garner, 1962; Garner & Morton, 1969).

so that $C(t)$ values exceed those on the right-hand side of the inequality violate the race model inequality. Similarly, the Grice's inequality can be written as,

$$C\,(t) \geq \frac{\log\{MIN\,[S_N\,(t)\,,S_S\,(t)]\}}{\log\,[S_N\,(t)\,S_S\,(t)]}$$

meaning that $C(t)$ values smaller than those on the right hand side violate the Grice's bound.

4.7. Capacity: Resilience functions

Recently, Little, Eidels, Fific, and Wang (2015) have considered the capacity predictions for situations in which distractor information is also present. The capacity coefficient defined earlier refers to the processing of double targets in relation to that of single targets when the latter appear without additional information (=distractors). However, in many cases a distractor can appear along with the to-be-detected target. Because the double-target condition does not entail distractor information, an advantage in processing might derive not only from increased efficiency, but also from the presence of distractors in the single target conditions. The resilience function developed by Little et al. (2015) addresses such situations, incorporating the possible influence of distractors. The development by Little et al. (2015) is useful especially when the distractors differ in salience: The resilience difference function then provides diagnostic information with respect to the underlying architecture.

Again, the diagnostic statistics is the resilience difference function $R_{\text{diff}}(t)$, which is the difference between two resilience functions, one with highly discriminable distractor and the second with a less discriminable distractor:

$$R_{\text{diff}}\,(t) = \frac{-\ln\,(S_{AB}\,(t))}{-\ln\,[S_{AH}\,(t)\,xS_{BH}\,(t)]} - \frac{-\ln\,(S_{AB}\,(t))}{-\ln\,[S_{AL}\,(t)\,xS_{BL}\,(t)]}$$

when $R_{\text{diff}}(t) = 0$ the model is parallel self-terminating, when $R_{\text{diff}}(t) < 0$ the model is either serial self-terminating, parallel exhaustive, or serial exhaustive, and when $R_{\text{diff}}(t) > 0$ the model is coactive (see Table 2 in Little et al., 2015).

Concerning the present study, the discrimination between number and size was trivially easy in which case, "the standard capacity coefficient...is...still useful" (Little et al., 2015, p. 29). Therefore, we decided to calculate both the original capacity functions as well as the resilience difference functions, especially as only the former applies in Experiment 1. Our Experiment 1 included only one type of distractor so that we could not calculate the *resilience difference* function for this experiment. We did calculate the function in Experiment 2 (to anticipate, this did not alter the results).

5. SFT interrogation of the SCE

The natural candidate architecture for generating the SCE is coactive processing. Automatic activation of numerical magnitude means that number information is always present along with physical size information (the relevant dimension in many studies) so that the two coactivate the response. Their interaction is expressed in the observed SCE. The coactivation can come in the form of coalescence into a common a-modal magnitude representation. Prominent theories assume the presence of such a representation, sustained by activity in the intra parietal sulcus (IPS; see e.g., Walsh, 2003; see also Dehaene, 2007; Piazza, Pinel, Bihan, & Dehaene, 2007). In this architecture, numerical magnitude and physical size coalesce into a single representation downstream. This architecture is consistent with the violation of the race model inequality and with supercapacity in processing (when magnitude and size correspond). In sum, the SCE is the behavioral manifestation of an underlying interactive architecture.

However, the absence of the SCE in many experiments casts doubt on the automaticity assumption. The approach advocated by Algom denies the assumption of automaticity in numerical processing (Namdar, Ganel, & Algom, 2018; Pansky & Algom, 1999, 2002). As an aside, Algom's approach is also at variance with the theory of a common magnitude code (Van Opstal & Verguts, 2013), a notion that is actually inconsistent with the main body of classic psychophysics (Marks, 1974; Marks & Algom, 1998; Stevens, 1975). All of this means that an architecture entailing coactivation or other forms of interaction is not absolutely obligatory.

The next plausible candidate architecture is the benchmark model of minimum time parallel independent race with unlimited capacity. In this architecture, there does not exist a cross-talk between number and size. The number horse does not know the position, pace, indeed the very existence of the size horse. Inevitably, the SCE, when it is present, does not derive from an interaction in processing. It can rather reflect a redundancy gain due to statistical facilitation (Raab, 1962; see Eidels, 2012; Eidels et al., 2010, for a similar theory with respect to the Stroop effect). For an exhaustive stopping rule, the fact that the system must await the processing of the slower signal might be taken as a weak form of dependence itself.

The least plausible candidate architectures for generating the SCE are serial models, whether minimum time or exhaustive. How can an interaction, a redundancy gain, or the SCE ensue from such processing? To a first glimpse, they seem impossible. After all, one signal is processed only after the other signal has completed processing, so that there is no overlap between the two. On this background, Townsend and Nozawa (1997) have nevertheless proved that a redundancy gain is still possible under a *serial* exhaustive architecture.

5.1. Predictions

Although an interactive architecture, in particular coactivation, is the model favored by the traditional view, it is problematic on two counts. First, the absence of the SCE in tightly controlled studies is inconsistent with the model. Second, the absence of an a-modal magnitude code detracts from the plausibility of coactivation.

Of the remaining models, both the benchmark parallel race model and some form of serial processing are possible. According to the former, the SCE, when it is present, derives from statistical facilitation in the face of redundant target. Consequently, this model also predicts relatively small amounts of SCEs.

Finally, we cannot exclude the possibility of a serial model. Following Townsend and Nozawa (1997), redundancy gain expressed as SCE is possible under serial processing. It is possible that one attribute is more salient or favorable to be processed first. Such an asymmetry leads quite naturally to serial processing.

To conclude, an SFT analysis, sustained by the DFP design, permits one to decide between the theoretical approaches suggested for an understanding of the phenomenon of the SCE.

5.2. A caveat with respect to the task

Let us issue a methodological caveat with respect to the task used. In the typical SCE study, the observers respond to a single dimension of the digit – its physical size – while attempting to ignore the task-irrelevant dimension of numerical value. In the current SFT paradigm, by contrast, the task was that of divided attention: The observer attended to *both* dimensions in order to report the presence of at least one target. Note however that the current task invites an interactive architecture more strongly than does the typical selective attention task. The instructions calling

on observers to attend to both dimensions virtually force an interaction — if there is one. Interactive processing is the natural result invited by the present task. Uncovering a non-interactive architecture would be all the more notable when viewed on the background of the environment created by the current task (cf. Eidels et al., 2010; Fitousi, 2016a; Townsend & Wenger, 2014).

6. The present study

In the present study, we made use of the rich arsenal of the SFT toolkit in the quest of specifying the system underlying a numerical processing. We were particularly interested to see whether the processing of numerical magnitude and physical size is interactive. In two experiments, the participants decided, while timed, whether at least one target was present in the display. In Experiment 1, we used the common, spatially integrated version, in which number and size were features of the same digit. In Experiment 2, we used a spatially separated version (Dormal & Pesenti, 2007; Fitousi & Algom, 2006), in which physical size was varied via the length of a line drawn beneath the digit (of a constant size). Using a different stimulus preparation increases the generalizability of our results. If the SCE is based on the mandatory interaction of number and size, a coactive architecture with supercapacity is expected. If the SCE is an optional effect, a parallel or a serial architecture is possible. The latter architectures are incompatible with the view that the SCE is a mandatory effect and that the effect results from the automatic activation of number.

7. Method

7.1. Participants

Eight young undergraduate students from Ariel University served as observers (4 females, mean age of 23.2), four in each experiment. All of the observers had normal or corrected-to-normal vision.

Experiment 1

Stimuli and Apparatus. The stimuli were the numbers, 1, 6, 9, presented in Times New Roman font, in a large (a) (1.41° × 0.90°), medium (b) (0.52° × 0.30°), or a small (c) (0.30° × 0.22°) format. Targets were defined as values larger than the boundary on numerical magnitude *or* on physical size. The number 5 served as the boundary to classify numerical magnitude, so that 6 and 9 were large values and 1 small value. In a similar fashion, the font size, 0.45° × 0.26°, served as the boundary for classifying size so that (a) and (b) were large values and (c) small value. For the "large" category, we used two values of numerical magnitude and two values of physical size to create the factorial design for the double target presentations. Clearly, there were two values of numerical magnitude and two values of physical size in the target-large category. Factorial combinations of these values created the factorial design for the double-target presentation. This salience manipulation was done with respect to the double-target stimuli only. Consequently, neither "1" in medium size nor "6" in small size were presented. A single value of number and a single value of size was used for the small non-target category. This tactic did not affect the outcome because we presented an equal number of large and small trials in the experiment, so that there was an equal probability of a YES and NO responses.

The stimuli were created via Microsoft Painter. The experiment was controlled by an IBM computer using Authorware software. The software run on an IBM-compatible computer and the stimuli were displayed on a VGA 15 in. color monitor set to a 1024 × 768 resolution.

Design and Procedure. Each observer performed in 4 consecutive days of testing. On each day, the observer completed 40 blocks of 32 trials, making for 1280 trials per day. Overall, each observer performed in 5120 trials. Each experimental block of 32 trials was partitioned into target- and non-target-trials in equal frequency, so that, again, there was an equal probability of a YES and a NO response.

The salience manipulation was carried out by independently increasing or decreasing the numerical or the physical distance from the respective boundaries (see Dehaene, 1997 or Moyer & Landauer, 1967 on the distance effect; see also Ashby & Maddox, 1994). Our goal here was to speed up processing of the numerical dimension alone by presenting a number (9) considerably larger than the boundary (5) (high salience, *H*), or to slow down processing by presenting a number (6) close to the boundary (low salience, *L*). In a similar manner, we could selectively speed up processing of physical size by presenting the stimulus in a very large format (a) (high salience, *H*), or to slow down the same processing by presenting a format (b) close to the boundary (low salience, *L*).

The participants were tested individually in a dark and quiet room. Testing on each day began with a 20 min dark adaptation. The viewing distance was 76 cm from the screen. Observers placed their chin on a chin rest. They were asked to give a YES response if the stimulus was larger than 5 in numerical magnitude, larger than the boundary format in physical size, or both. The observer was to respond NO if the stimulus was smaller than the boundary in both numerical magnitude and physical size. Each experimental trial started with a blank screen presented for 500 ms, followed by the stimulus. The observer's response terminated the trial. The next stimulus appeared 350 ms after response. In a practice session prior to the main experiment a numeral at the criterion was repeatedly shown to the participant.

The responses were manual, performed by pressing the appropriate key on the computer board. The participant pressed the key 'm' for a YES response and pressed the key 'z' for a NO response. Participants were asked to respond as fast as possible but without making errors.

Experiment 2

Stimuli and Apparatus. The stimuli consisted of the numbers 8 and 2, appearing in the constant physical size of 0.6° × 0.37° (printed in "Ariel" font). A thin straight line was drawn beneath each number. The short line subtended 0.52° of visual angle, whereas the long line subtended 1.13° of visual angle. Both lines were of the same width (0.09°). The line appeared 0.07° below the number. In a congruent display, the large number 8 was positioned above the long line or the small number 2 was positioned above the short line. In an incongruent display, the large number 8 was positioned above the short line or the small number 2 was positioned over a long line.

The salience manipulation in this experiment was carried out by independently increasing or decreasing the brightness of the numbers and the lines. Thus, we could speed up processing of the numerical dimension alone by presenting the number in a very bright white color (high salience, *H*), or slow down processing by presenting the same number in a dimmer white color (low salience, *L*). In a similar manner, we could speed up performance with the physical dimension by presenting the line in a bright white color (high salience, *H*), or slow down performance by presenting the number in a dimmer white color (low salience, *L*). Clearly, the salience manipulation in Experiment 2 entailed all combinations.

7.2. Procedure

The participants were asked to give a YES response if the presented number was larger than 5, the line beneath it was the long one, or both, and a NO response if the number was smaller

Table 2
Mean RT (in ms) and percent error for double-target, single-target, and no-target displays in two experiments (H = high salience, L = low salience, with left-hand letter standing for number). The first number in each entry for an observer denotes the RT and the second number below the RT denotes the pertinent percent error.

	Observer	Double				Single		No Target
		HH	HL	LH	LL	Number	Size	
Exp. 1	1	368	393	379	404	437	406	410
		4.4	4.8	5.64	6.3	4.0	6.3	4.7
	2	352	366	381	394	434	393	392
		1.8	2.4	3.6	4.8	17.0	13.6	6.0
	3	322	332	336	348	393	376	332
		2.4	3.03	3.3	3.0	11.2	11.4	6.5
	4	388	425	412	456	468	467	431
		7.3	3.6	2.4	6.0	4.2	24.3	7.2
Exp. 2	1	440	452	460	478	489	544	473
		3.6	6.9	2.7	4.2	5.4	21.4	5.9
	2	389	404	399	411	429	452	397
		5.4	6.6	10.6	9.3	21.1	23.1	11.6
	3	502	519	519	528	656	627	551
		2.7	2.4	2.7	3.3	12.1	13.1	5.2
	4	444	453	463	472	485	563	459
		3.3	5.5	4.3	4.1	24.1	13.3	4.1

than 5 and the line beneath it was the short one. For the double-target presentations, all four combinations of the brightness of the number and the brightness of the line created the factorial design. In Experiment 2, too, the a priori probability of the YES and NO responses was equal.

All the other details of testing, design, and procedure in Experiment 2 were the same as in Experiment 1.

8. Results

The primary data of interest involved (a) the double (redundant) target trials that served to uncover the underlying architecture, and (b) the one- and two-target trials enabling the capacity analyses. The mean RTs for each observer in each experiment are listed in Table 2. Error rates amounted to 4.7%, 7.3%, 6.9%, and 9.4% for the 4 observers in Experiment 1, and 7.5%, 14.4%, 6.5%, 7.9% for the 4 observers in Experiment 2. The RT analyses are confined to correct responses only (RTs less than 200 ms or greater 1500 ms were also excluded, 1.3% overall).

8.1. Primary RT and error data

The results of Table 2 exhibit consistent advantage for double-target over single-target displays. The significance of this advantage is examined and discussed in a subsequent section. Another notable feature of Table 2 is the fairly uniform distribution of error rates in the double-target conditions. In particular, error rates do not increase in the LL condition, excluding the possibility of concealing an underlying coactive signature (cf. Fific et al., 2008). In Table 3, we present the results of the analysis on the primary data of interest: double-targets with varying levels of salience along each dimension.

The results of the ANOVA are consistent across all observers. The main effects of physical size and numerical magnitude were highly significant. However, the most revealing feature of the data is the lack of a size x number interaction. To a first approximation, these results indicate the absence of a cross-talk between physical size and numerical magnitude in processing.

Table 3
Results of ANOVA for each of 8 individual observers in two experiments.

	Source	MSE	F	p
Exp.1	Size	149 285	46.36	0.000***
	Number	27 475	8.53	0.003***
	Size × Number	96	0.00	0.86
	Size	249 773	58.63	0.000***
	Number	50 955	11.96	0.000***
	Size × Number	583	0.13	0.8
	Size	68 645	23.89	0.000***
	Number	40 593	14.13	0.000***
	Size × Number	396	0.13	0.71
	Size	225 249	18.75	0.000***
	Number	520 154	43.31	0.003**
	Size × Number	4434	0.36	0.54
Exp.2	Size	89 805	7.49	0.006**
	Number	166 103	13.86	0.000***
	Size × Number	907	0.07	0.78
	Size	44 951	9.89	0.002**
	Number	18 253	4.01	0.04*
	Size × Number	377	0.08	0.77
	Size	52 599	5.74	0.01*
	Number	40 593	4.86	0.02*
	Size × Number	7605	0.83	0.36
	Size	27 584	4.38	0.036*
	Number	122 542	19.49	0.000***
	Size × Number	666	0.10	0.74

* $p < 0.05$.
** $p < 0.01$.
*** $p < 0.001$.

The data summarized in Table 3 are consistent in showing the absence a number x size interaction. However, the lack of evidence is not the same as positive evidence for the lack of interaction. Therefore, in Table 4, we quantify the evidence for the null hypothesis using Bayesian analysis. A glimpse, at Table 4 shows that this evidence is strong. In Table 4, we also provide the results aggregated over the individual observers in each experiment. The outcome of the aggregate analysis provides further support for the hypothesis of lack of interaction.

Table 4
Bayesian ANOVAs, for 8 individual observers and for the aggregate data of Experiments 1 and 2.

	Obs.	Source	P (incl)	P (incl|*data*)	$BF_{inclusion}$
	1	Size	0.6	1.00	>**155**
		Number	0.6	0.84	**3.56**
		Size × Number	0.2	0.07	0.33
	2	Size	0.6	1.00	>**155**
		Number	0.6	0.96	**15.92**
		Size × Number	0.2	0.09	0.32
Exp.1	3	Size	0.6	1.00	>**155**
		Number	0.6	0.98	**48.43**
		Size × Number	0.2	0.08	0.35
	4	Size	0.6	0.99	>**155**
		Number	0.6	1.00	>**155**
		Size × Number	0.2	0.09	0.43
	Aggregate	Size	0.6	1.00	>**155**
		Number	0.6	1.00	>**155**
		Size × Number	0.2	0.05	0.19
	1	Size	0.6	0.74	**1.90**
		Number	0.6	0.98	**41.80**
		Size × Number	0.2	0.06	0.26
	2	Size	0.6	0.89	**5.71**
		Number	0.6	0.78	**1.02**
		Size × Number	0.2	0.02	0.1
Exp.2	3	Size	0.6	0.65	**1.23**
		Number	0.6	0.66	**1.31**
		Size × Number	0.2	0.05	0.21
	4	Size	0.6	0.72	**1.56**
		Number	0.6	1.00	>**155**
		Size × Number	0.2	0.36	0.54
	Aggregate	Size	0.6	0.99	>**155**
		Number	0.6	1.00	>**155**
		Size × Number	0.2	0.05	0.20

Table 5
One sided Kolmogorov–Smirnov D^ values on the survival functions. Significant values entail that H_0 was refuted and ordering held. Note that the relations between S_{LH} and S_{HL} were not tested because their relations have no influence on ordering. [See text for H, L, and S].

Observer	Experiment 1					Experiment 2				
	$S_{LL} > S_{HH}$	$S_{LL} > S_{HL}$	$S_{LL} > S_{LH}$	$S_{LH} > S_{HH}$,	$S_{HL} > S_{HH}$	$S_{LL} > S_{HH}$	$S_{LL} > S_{HL}$	$S_{LL} > S_{LH}$	$S_{LH} > S_{HH}$,	$S_{HL} > S_{HH}$
1	0.32***	0.25***	0.11*	0.01*	0.25***	0.19***	0.13**	0.12*	0.10*	0.11*
2	0.31***	0.27***	0.10*	0.22***	0.13*	0.18***	0.09	0.11*	0.11*	0.16***
3	0.32***	0.25***	0.12**	0.14***	0.24***	0.19***	0.08	0.09	0.13*	0.13*
4	0.27***	0.23***	0.16**	0.14***	0.11*	0.28***	0.18***	0.13**	0.18***	0.14***

* $p < 0.05$.
** $p < 0.01$.
*** $p < 0.001$.

8.2. Mean-based analyses

The MICs for the subset of the redundant target trials for each of the individual observers are given in Fig. 5. A glimpse at Fig. 5 reveals that the forms of the functions are highly consistent across all observers. The functions are separated by a constant amount along the vertical dimension in each panel, their parallelism attests to serial processing of either minimum time or exhaustive mode. Our results argue against any form of interactive processing of either over additive or coactive variety.

8.3. Distribution-based analyses

To enable further specification of the underlying architecture and stopping rule, we calculated the survivor interaction contrast function for each observer. Fig. 6 exhibits these SIC(t) functions. The SIC functions are remarkably consistent across observers: All hover at around zero. This finding is consistent with serial minimum time self-terminating processing. To be more precise, it is a serial, minimum time process that implies both a flat SIC function, and independence.

Statistical inference with respect to the SIC functions presented in Fig. 6 was based on a test developed by Houpt and Townsend (2010). The procedure entails two separate null hypothesis tests: One examines whether the largest positive value of the SIC is significantly different from zero and the second examines whether the largest negative value (in an absolute sense) is significantly different from zero. None of the former (D^+; with D^ the supremum of these differences, see Houpt & Burns, 2017, p. 57) turned out significant, whereas all but one of the latter (D^-) were significant (values 0.34, 0.34, 0.43, 0.12 [Experiment 1], and 0.21, 0.27, 0.26, and 0.31 [Experiment 2]. This outcome provides further evidence for the absence of coactivation; the result is inconsistent with parallel models (including coactivation) as well as with serial exhaustive models.

8.4. Selective influence: Ordering of the survivor functions

To validate our manipulations and analyses, we calculated the survivor functions for the four redundant target trials. Satisfying their predicted ordering is a necessary (though not sufficient) condition of exhibiting selective influence. Fig. 7 shows the four survivor functions for each observer. Observe that the predicted ordering, HH > LH, HL > LL, holds well for each of the individual observers, supporting the presence of selective influence in our data. Table 5 presents the outcome of Kolmogorov–Smirnov tests performed on these survivor functions. The results summarized in Table 5 further support selective influence.

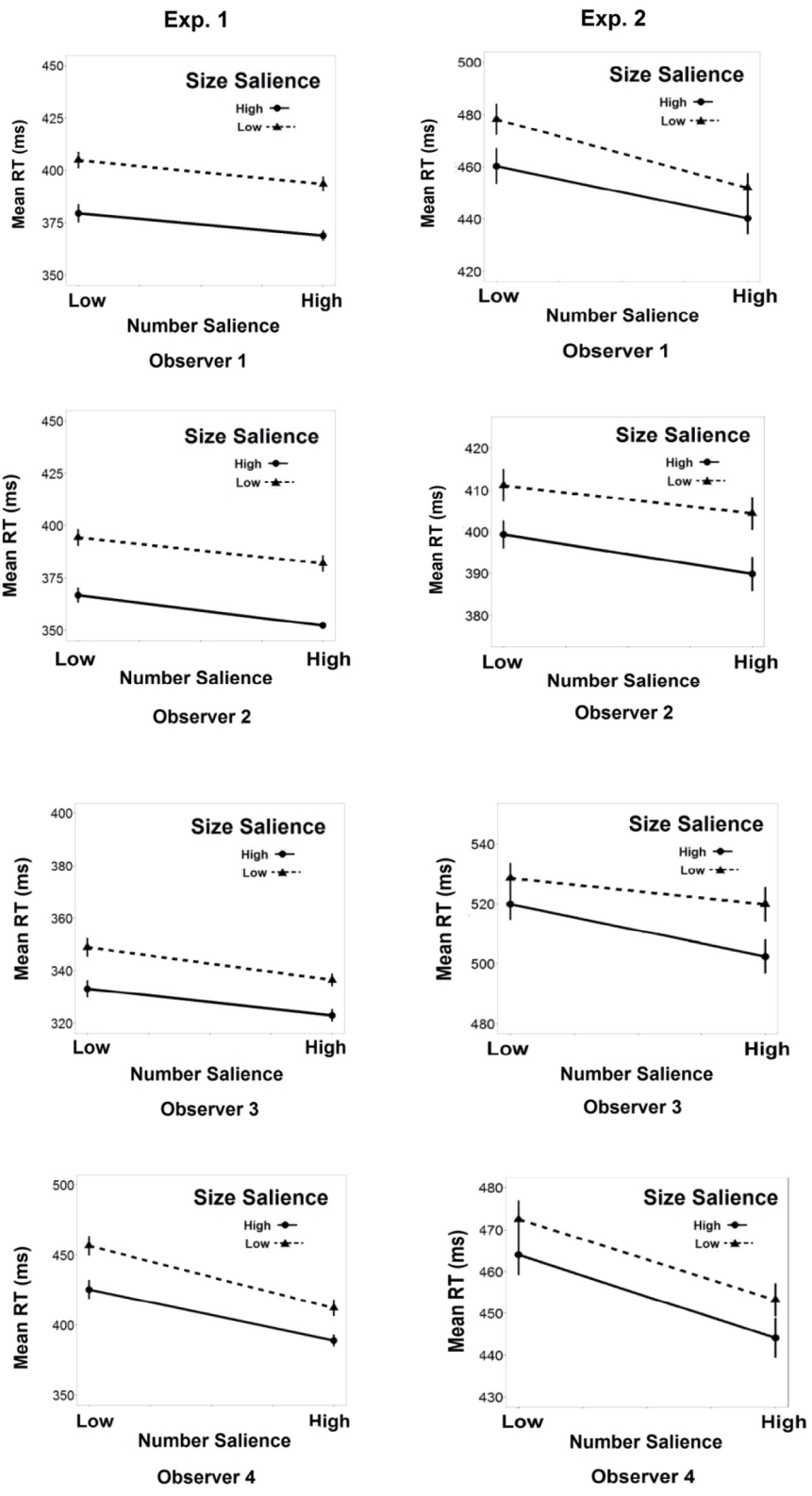


Fig. 5. MIC for 8 individual observers in Experiments 1 and 2.

8.5. Measuring capacity

We derived the capacity coefficient, $C(t)$, the Miller inequality, and the Grice inequality, separately for each observer. Applying the routine developed by Townsend and Eidels (2011), the three capacity measures appear within a single unified space (Fig. 8). Observe that although $C(t)$ hovers at around unity, capacity is a bit strained under most values of t. Inspection of Fig. 8 shows that the system is unlimited to slightly limited capacity at all time t. As a rule, the Grice bound is not violated and the Miller bound is

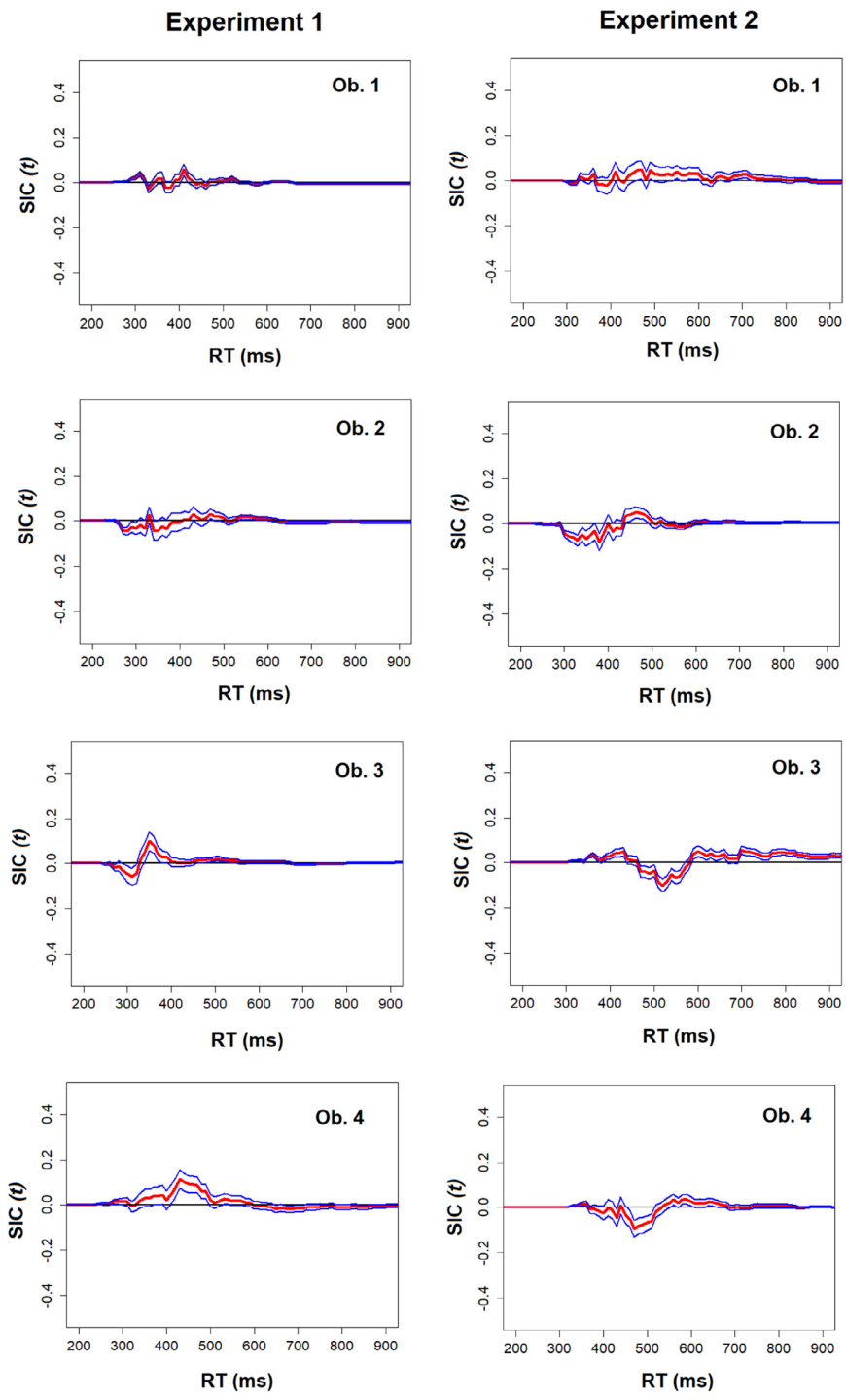


Fig. 6. SIC(t) contrasts for each of 8 observers in the two experiments.

generally also not violated, although Fig. 8 reveals some violations (in 2 of the 8 participants). Overall, visual inspection joins the results of the statistical test ruling out supercapacity.

Both Experiments 1 and 2 included distractor information, so analysis by the difference resilience function is invited (Little et al., 2015). However, two points are notable. First, difference resilience analysis is not feasible with Experiment 1 due to the fact that we used only a single value of the distractor. Second, the distractor dimension in our experiments was eminently discriminable (not one observer was ever confused over number and physical size, a condition in which resilience analysis gives the same outcome as standard capacity analysis). Indeed, as Fig. 9 shows, the difference resilience analysis of the data of Experiment 2 indicates the absence of supercapacity, inconsistent with coactive processing.

A recent test developed by Houpt and Townsend (2012) compares the observed RTs with those predicted by an unlimited capacity independent parallel model. In Experiment 1, the z cores were negative: −25.9, −26.2, −25.85, and −19.94 for observers 1 to 4 respectively ($p < .005$ in all cases). In Experiment 2, the z cores were also negative: −15.0, −16.6, −23.1, and −14.75 for observers 1 to 4 respectively (again $p < .005$ in all cases). These results

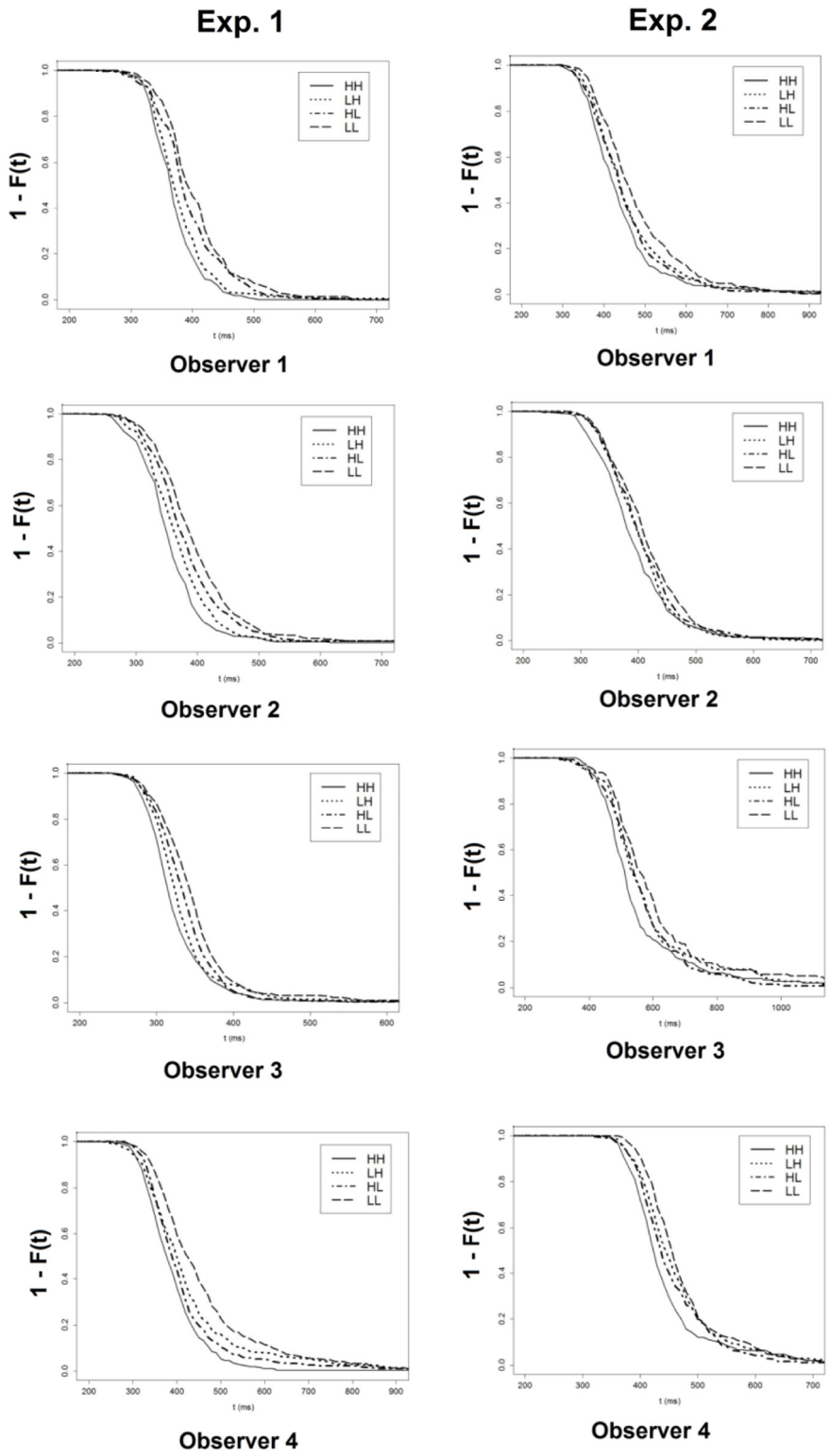


Fig. 7. Estimated survivor functions for 8 observers from the two experiments in the redundant targets condition under the factorial salience manipulations.

provide strong evidence for $C(t)$ that is smaller than 1, hence for limited capacity.

In their original work, Little et al. (2015) note that the $C(t)$ is still useful when target and distractor dimensions are highly discriminable, as they are in our case. A possible implication is that the distractor does not affect target processing (i.e., context invariance is in force). We believe that our data are consistent with zero effect of distractors on target processing. Nevertheless, for a further direct test of the possible impact of distractors, we considered the target-distractor presentations in Experiment 2 (Table 2). For each target, the distractor could be highly salient (H) or less highly salient (L). We compared the two classes of trials for both RT and error by the very sensitive Bayesian BM test for each of the 4 observers. Of the 16 comparisons, 13 yielded a BM value

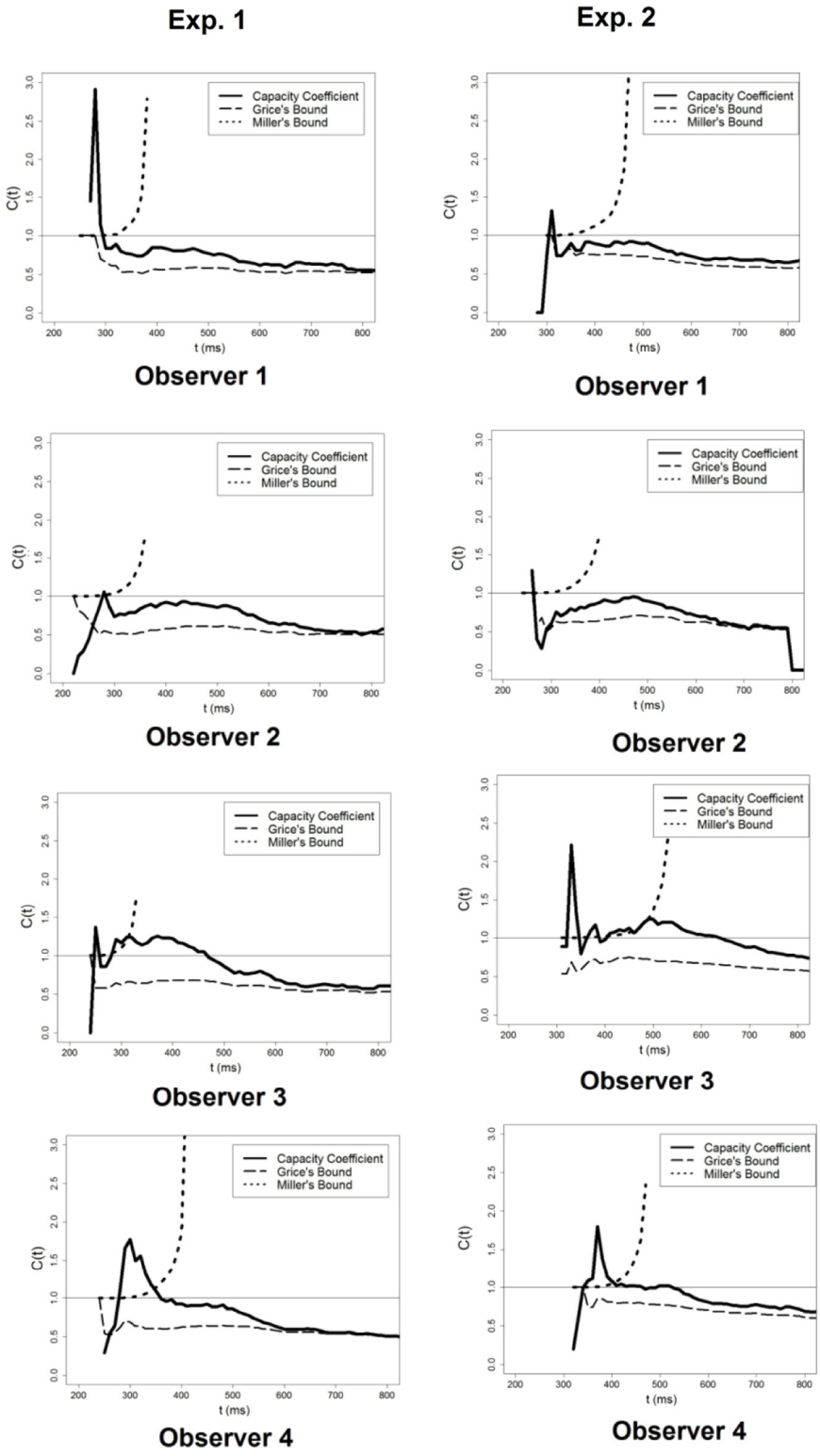


Fig. 8. Capacity coefficient space for each of 8 observers in two experiments.

of 1 or less. Therefore, there is insubstantial evidence to support the hypothesis that the salience of the distractor affected target performance.

For a final test on any residual effect of distractor information, we recruited a new observer who performed in Experiment 2, but without distractors. The design was exactly the same as that used with our 4 observers, but now only double targets, single-targets (or no-targets) were presented. To reiterate, the single targets were presented alone. The capacity results of our new observer are shown in Fig. 10. Note that in the absence of distractors, $C(t)$ is the relevant capacity measure (the unified space includes the Grice and the Miller's bounds as well). What is most remarkable about the results of the new observer is their close affinity to the results of the observers of Experiment 2, who performed with distractors.

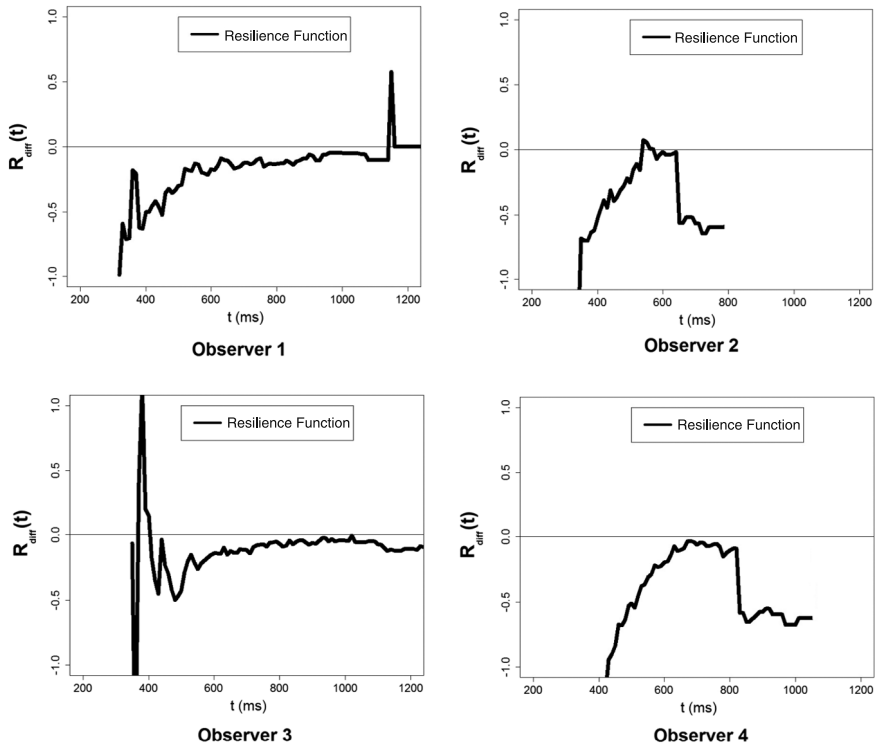


Fig. 9. Difference resilience functions for the 4 observers in Experiment 2. Note that all the functions are below zero for most values of t consistent with a serial processing.

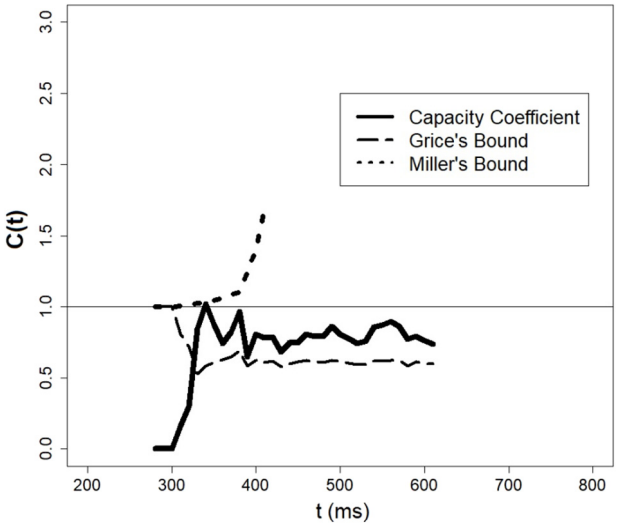


Fig. 10. Capacity space for 1 observer in the version of Experiment 2 without distractors.

All data revealed limited (to-unlimited) capacity, and no violation of the race model inequality. The collective data is consistent with the conclusion that the distractors did not have an impact on target processing in our study.

8.6. Miller's inequality and redundancy gains

On the background of satisfying Miller's inequality (see Fig. 8 again) in tandem with the lack of supercapacity, it may come as a surprise that we have recorded appreciable amounts of redundancy gains with all observers. All of our observers responded faster to double-target displays than to single-target displays (the gains ranged between 15 and 89 ms and all were statistically significant at $p < 0.01$). Note that the double-target stimuli were also size–magnitude congruent, whereas the single-target stimuli were size–magnitude incongruent. Therefore, the redundancy gains recorded are simultaneously those of the SCE. Given the present results of (a) minimum-time, serial architecture, (b) the satisfaction of the Miller inequality, and (c) the absence of supercapacity, it is clear that the SCE cannot derive from any form of an interaction between physical size and numerical magnitude.

9. Discussion

Given the central role of the SCE in numerical cognition, specifying the attendant architecture is important. According to accepted wisdom, the interference to judgments of physical size by numerical value (=SCE) results from the interaction in processing between these two dimensions of the numeral. Moreover, the interaction is mandatory due to the alleged automaticity in processing numerical magnitude. The present results argue against this view. We found the architecture to be that of serial processing with a minimum-time stopping rule. This outcome excludes number–size interaction. Our results support instead an independent processing of number and size.

Surprise: Redundancy Gains are Mandatory under Certain Versions of Serial Processing

In the face of the present serial minimum-time model, we still found appreciable amounts of redundancy gain — speedier responses to double-target than to single-target presentations. Note that this redundancy gain is actually an SCE. On a first glimpse, the pattern is puzzling: How is it possible to record a significant redundant target effect within a serial minimum-time model? A ready solution involves the concept of context invariance (Colonius, 1990; Luce, 1986; Townsend & Wenger, 2004). Context invariance holds if processing of a target remains the same regardless of whether the target is presented alone or jointly with another target. However, in the present context a single target is

never presented alone, but jointly with a distractor. Therefore, we assume stochastic independence to hold between the single target and the distractor (as well as between the two targets). Then, it is easily shown that the present serial model not only allows for redundancy gains but also that it actually *mandates* such gains. Consider the completion times, T, in the present serial minimum-time model with targets A and B, assuming that the two distractors, X and Y, are processed at an equal rate V, so that $V_x = V_y = V$:

Double-Target condition with Targets A and B: $T = E(T_A, T_B)$

Single-Target condition with Target A and Distractor X: $T = E(T_A, T_x)$

Single-Target condition with Target B and Distractor Y: $T = E(T_B, T_y)$.

Given a minimum-time serial model with a random order of channel processing,

$$E(T_A, T_B) < E(T_A, T_x)$$

and,

$$E(T_A, T_B) < E(T_B, T_y),$$

each of which denotes the redundant target effect. The effect derives from the time consumed by the distractor when it is contained in the channel that is processed first. If we further assume that the rate of processing is equal (stochastically) across the two targets, then the two inequalities are interchangeable and collapse into a single measure of redundancy gain. Note, too, that the poorer the processing of the distractor with respect to the target, e.g., $T_x \gg T_A$, the larger the redundant target effect.

Consider fixed-order serial models, focusing on the target-distractor displays. When the channel containing the distractor is processed first, the inequalities remain unchanged so that double target displays are faster than single target plus distractor displays. However, if order is fixed such that the channel containing the single target is processed first, the redundant target effect is no longer present. This outcome is of great importance for understanding the variability in the literature with respect to the presence and magnitude of the SCE. As we recounted, the SCE is often present, but it is absent in a fair amount of cases. Our results show that when the SCE is present, the underlying architecture is serial minimum time of mixed-order or of a specific fixed-order. Order is fixed, so that the distractor is processed first in target-distractor displays. In contrast, in studies in which the SCE is absent, the underlying architecture is serial minimum time, but now the channel containing the single-target is processed first in target-distractor displays.

Can one predict the precise species of the serial minimum time model used and, consequently, the appearance of the SCE? We argue that prediction is possible given that processing is governed by the contextual factor of relative dimensional salience discussed earlier (Fitousi & Algom, 2006; Melara & Algom, 2003). The SCE is bound to appear when the task irrelevant dimension of numerical magnitude is more salient than the task relevant dimension of physical size. This was the case in the great majority of published SCE studies. Note that enhanced salience increases the probability for distractor number to be processed first. As we just showed, this order must yield a SCE. When number and size are equally salient, the SCE might or might not appear. This setup is consistent with a mixed-order architecture that sometimes favors the distractor and sometimes favors the target in priority. As Algom and his associates have shown, the SCE is not present in cases in which the to-be-ignored dimension, is less salient than the target dimension (see Fig. 1, again). This last setup favors priority for target processing in target-distractor displays, a fixed-order that does not yield the SCE.

All of the current analyses converge on the conclusion that the underlying model is serial minimum-time, of a mixed-order or of a specific fixed-order variant. The analyses included also capacity with $C(t)$ and the difference $R(t)$ measures. As we show, the redundant target effect, when it is observed, is a natural derivative of the underlying serial model.

Possible Violations of the Race-Model Inequality under Exhaustive Processing

Our data are consistent with a serial minimum-time model. The data also entailed appreciable amounts of redundant target advantage, but did not in general violate the race model inequality. On this background, it seems instructive to take a look at serial exhaustive models. Townsend and Nozawa (1997) have provided a proof that such models can violate the race model inequality. Again, two different features of the current situation should be kept in mind. First, we found self-terminating not exhaustive processing, and, second, the redundancy gains found, while appreciable, did not violate the race model inequality. With these points in mind, the development detailed in Townsend and Nozawa (1997), pp. 598–599) for an exhaustive stopping rule can be applied virtually unchanged here. Their proof enjoys generality (subject to the perfunctory change in a parameter due to the alternative stopping rule). Given the proof by Townsend and Nozawa (1997), it follows immediately that *with stochastically independent, serial minimum-time processing, and with context independence in force, there exist distributions with stochastically faster times to targets than to non-targets in which violations of the race-model inequality are possible*. Note that our results are a bit more conservative as we did not even witness violations of the inequality.

In conclusion, the results show that the SCEs observed in our study do not derive from an interaction between number and size. Serial models exclude the possibility of such an interaction. Simultaneously, our results challenge the theory of automatic activation of numerical magnitude. Such activation mandates interaction in processing. Again however, interaction was absent in our results. We note that the authors of a recent study using other methods (Sobel, Puri, Faulkenberry, & Dague, 2017) have similarly reported independent processing of the dimensions of number–size stimuli. The results of the recent study by Namdar et al. (2018) cast serious doubt on the automaticity account of the SCE. Therefore, the present results show that the SCE is not obligatory. The contextual factors identified by Algom and Fitousi arguably determine the model and the priority of processing. (e.g., Algom & Fitousi, 2016; Algom et al., 2017; Fitousi, 2014, 2016b; Fitousi & Algom, 2006; Fitousi et al., 2009; Namdar et al., 2018).

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